



Innovation Healthcare Private Limited

Title

**Summary of Safety and Clinical Performance (SSCP) of Everolimus
Eluting Coronary Stent System**



Innovation Healthcare Private Limited

Manufacturing Site Address:

Plot No. 105, Solitaire Industrial Park, Village Dehmikalan, The. Sanganer, Jaipur – 303007, Rajasthan (India)

Document Title: **Summary of Safety and Clinical Performance (SSCP) of Everolimus Eluting Coronary Stent System**

Brand Name(s): **Eternia, Eternia Select, Eternia Brio, Eternia Nexa, Eternia AB**

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This Summary of Safety and Clinical Performance (SSCP) is intended to provide public access to an updated summary of the main aspects of the safety and clinical performance of the device. The SSCP is not intended to replace the Instructions for Use as the main document to ensure the safe use of the device, nor is it intended to provide diagnostic or therapeutic suggestions to intended users or patients.



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1 Device identification and general information

1.1 Device trade name

Drug Eluting Coronary Stent System

1.2 Manufacturer's name and address

Name: Innovation Healthcare Pvt. Ltd.

Manufacturing Site Address: Plot No. 105, Solitaire Industrial Park, Village Dehmikalan, The. Sanganer, Jaipur – 303007, Rajasthan (India)

1.3 Manufacturer's single registration number (SRN)

IN-MF-000042888

1.4 Basic UDI-DI

890611210EECSBA

1.5 Medical device nomenclature description / text

Applicable EMDN code: P070402010301 – Coronary Drug Eluting Stents (DES) with Polymers

1.6 Class of device

The Everolimus Eluting Coronary Stent System is classified as Class III device according to Rule 14 (Special rule – All devices incorporating, as an integral part, a substance which, if used separately, can be considered to be a medicinal product) of Chapter III, Annex VIII, and Classification Rules of MDR 2017/745.

1.7 Year when the first certificate (CE) was issued covering the device

NA

1.8 Authorised representative; name and the SRN

CMC Medical Devices & Drugs SL

SRN: ES-AR-000000293

1.9 NB's name and the NB's identification number

Name of Notified Body: Polish Centre for Testing and Certification S.A. (PCBC) (1434)

Address: ul. Puławska 469, 02-844 Warsaw, Poland.

2 Intended use of the device

2.1 Intended purpose and Indications

Everolimus Eluting Coronary Stent system (EECS) is intended for use in the patient eligible for Percutaneous Transluminal Coronary Angioplasty (PTCA). Everolimus Eluting Coronary Stent System is indicated for improving coronary luminal diameter in patients with symptomatic ischemic disease due to discrete *de novo* lesions of length ≤ 48 mm in native coronary arteries with a reference vessel diameter of 2.00 to 4.50 mm.

2.2 Target population(s)

- Patient who are eligible for Percutaneous Transluminal Coronary Angioplasty (PTCA).
- Patients with symptomatic ischemic disease due to discrete *de novo* lesions of length ≤ 47 mm in native coronary arteries with a reference vessel diameter of 2.00 to 4.50 mm.
- Patient with age 18 years and above.
- Patients who do not have any contraindications for using the product.

2.3 Contraindications and/or limitations

Use of Everolimus eluting coronary stent system is contraindicated in the following patient types:

- Patients with hypersensitivity or allergies to aspirin, heparin, Clopidogrel, ticlopidine, drug such as Everolimus or similar drugs on analogue or derivative, cobalt, chromium, nickel, molybdenum, tungsten or any contrast medium.
- Patients in whom, anti-platelet and/or anticoagulant therapy are contraindicated.
- Patients judged to have a lesion that prevents complete inflation of an angioplasty balloon.
- Transplant Patients.

3 Device description

3.1 Description of the device

Everolimus eluting coronary stent system uses an Everolimus drug and biodegradable polymer coated on the cobalt chromium stent platform. The cobalt-chromium platform of this device offers very thin strut

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thickness and decent radiopacity during the PTCA (Percutaneous Transluminal Coronary Angioplasty). The stent has an open cell design and features good flexibility, crossability and conformability. The coating process allows the integrity of coated polymer after excessive stent expansion and a systemic elution of Everolimus in a well -controlled fashion after the stent implantation.

Everolimus Eluting Coronary Stent is mounted on a folded balloon attached to a catheter delivery system for the placement into a coronary artery (blood vessel supplying blood to the heart). The Everolimus Eluting Coronary Stent System is designed to treat coronary arteries with a diameter from 2.00 mm to 4.50 mm with stent length varying from 8 mm to 52 mm.

The drug content on the stent is in constant quantity of drug per square area of stent; however, as the stent length varies from 8 mm to 52 mm, the total drug loading on each stent varies.

3.2 A reference to previous generation(s) or variants if such exist, and a description of the differences

No previous generation of the device have been brought on the market by Innovation Healthcare Pvt. Ltd..

3.3 Description of any accessories which are intended to be used in combination with the device

No accessories for Everolimus Eluting Coronary Stent system are identified.

3.4 Description of any other devices and products which are intended to be used in combination with the device

Everolimus Eluting Coronary Stent System is intended to be used with the following devices:

- Guiding catheter
- Guide Wire
- Inflation device
- Y Connector

4 Risks and warnings

4.1 Residual risks and undesirable effects

There are no remaining residual risks associated with the use of the device.

Undesirable effects

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Undesirable effects / potential adverse events which may be associated with the use of a stent in native coronary arteries include but are not limited to:

- Abrupt stent closure.
- Acute myocardial infarction.
- Allergic reactions to anti – coagulant and or antithrombotic therapy or contrast medium.
- Angina.
- Aneurysm
- Arrhythmia, including ventricular fibrillation (VF) and ventricular tachycardia (VT).
- Arterial perforation.
- Arterial rupture.
- Arteriovenous fistula.
- Bleeding complications.
- Bradycardia.
- Cardio tamponade.
- Cardiogenic shock.
- Coronary spasm
- Coronary or stent embolism.
- Coronary or stent thrombosis.
- Death.
- Dissection of the coronary artery.
- Emboli, distal (air, tissue or thrombotic emboli)
- Emergency or non – emergent Coronary Artery Bypass Graft Surgery.
- Entry site complications.
- Heart Failure.
- Hematoma.
- Hemorrhage, requiring transfusion.
- Hypotension / Hypertension.

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- Infection.
- Infection and / or pain at the access site.
- Injury to the coronary artery.
- Ischemia.
- Nausea and vomiting.
- Palpitations.
- Perforation or rupture.
- Pericardial effusion. Pseudoaneurysm, femoral.
- Renal failure.
- Respiratory failure.
- Restenosis of stented segment.
- Rhythmical disturbances.
- Shock / Pulmonary edema.
- Spasm.
- Stroke / cerebrovascular accident / TIA.
- Total occlusion of coronary artery.
- Unstable angina pectoris.
- Vascular complications, which may require vessel repair.
- Ventricular fibrillation.

4.2 Warnings and precautions

- Please ensure that the inner package has not been opened or damaged as this may indicate the sterile barrier has been breached.
- Do not Re-sterilize and/or reuse this device. Reuse or Re-sterilization may create a risk of contamination of the device and/or cause patient infection or cross infection, including, but not limited to, the transmission of infectious disease(s) from one patient to another and it may also compromise the structural integrity of the device. The device is intended for single use only.

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- Should unusual resistance be felt at any time during lesion access or delivery system removal, the entire guiding catheter and stent system should be removed as a single unit.
- Applying excessive force to the stent delivery system can potentially result in loss or damage to the stent and delivery system components.
- Since the use of this device carries the associated risk of sub-acute thrombosis, vascular complications and / or bleeding events, judicious selection of patients is necessary.
- Patients with known allergy to Cobalt alloy (L605) may suffer an allergic reaction to this implant, and the risk benefit ratio should be carefully assessed.
- The device should be manipulated while under high – quality fluoroscopic observation. Do not advance or retract the catheter unless the balloon is fully deflated under vacuum. If resistance is met during manipulation, determine the cause of resistance before proceeding.
- Balloon pressure should not exceed the rated burst pressure. The rated burst pressure is based on the results of in vitro testing. Use of a pressure monitoring device is recommended to prevent over pressurization.
- Device is to be used under X-rays fluoroscopy, hence standard method for using this imaging technique needs to be followed.
- Use the device before the “Product Used by date” specified on the package

4.3 Other relevant aspects of safety, including a summary of any field safety corrective action (FSCA including FSN) if applicable

There has not been any Field Safety Corrective Action (FSCA) or Field Safety Notice (FSN) for Everolimus Eluting Coronary Stent System.

5 Summary of clinical evaluation and post-market clinical follow-up (PMCF)

5.1 Summary of clinical data related to equivalent device, if applicable

Current clinical data available is based on Everolimus Eluting Coronary Stent System

5.2 Summary of clinical data from conducted investigations of the device before the CE-marking, if applicable

NA. The device is not yet CE marked.

5.3 Summary of clinical data from other sources, if applicable

Clinical data exists from a variety of sources, including the literature search and analysis for similar devices and completed Post Market Clinical Follow – up (PMCF) study of subject device. It is to be noted that the majority of the clinical data comes from the PMCF study of the device, however for building confidence for the safety and performance of the device, clinical data of similar are considered. These clinical data are utilized in Clinical Evaluation Report and these are summarized below:

5.3.1 Literature Review

The subject device is a Everolimus Eluting Coronary Stent System. This device is manufactured by using long term established manufacturing technology. It is a state-of-art device and it is performing well in the real life patients as proved by customer feedbacks listed in the PSUR. The complaints were identified in risks analysis and were mitigated to the acceptable levels.

The results of invitro and pre-clinical performances tests give clear indications that the concern devices pass all the tests criteria and it complies with the requirements of the applicable product standards.

The safety and performance of the device is further supported by the clinical data obtained from literature search:

- A. Several studies demonstrated that the bioabsorbable polymer drug eluting stent shows better clinical outcomes than the durable/permanent polymer drug eluting stents including financial benefits. A bioresorbable polymer sirolimus-eluting stent with lower peri-procedural myocardial infarction (ppMI) rates results in lower direct medical expenses while also enhancing patient outcomes. The first hospitalization accounts for the majority of the cost savings, with only modest savings up to 12 months after discharge¹. Regarding 1-year target lesion failure, another randomized trial found that the SYNERGY bioabsorbable polymer everolimus-eluting stent was noninferior to the PROMUS Element Plus everolimus-eluting stent. These data demonstrate that SYNERGY is generally safe and effective for a variety of patients undergoing percutaneous coronary intervention². Further, in a

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sizable, randomized trial, patients receiving (Biodegradable Polymer Sirolimus Eluting Stents) BP SES as opposed to DP EES (Durable Polymer Everolimus Stents) had considerably lower rates of target lesion failure as well as target artery MI, clinically driven target lesion revascularization, and late or very late stent thrombosis after 3 years. The findings confirm that ultrathin-strut BP SES continues to be better to DP EES³. Studies also proves that BP-SES with ultra-thin struts is both safer and more effective than DP-EES across the board, especially in small vessels⁴. Significant differences in TLF and target vessel-related MI maintained in a large, randomized trial for 2 years, favoring BPSES therapy versus DPEES. BPSES was also found to significantly reduce late/very late stent thrombosis and cumulative target lesion revascularization⁵. However, studies showed that in calcified lesions, both the bioresorbable-polymer sirolimus-eluting stent and the durable-polymer everolimus-eluting stent were equally efficient and secure⁶ but BP- SES demonstrated a tendency toward a lower TLF rate than DP-EES for calcified lesions ⁷. In the study of MiStent Vs Xience (i.e biodegradable polymer Vs durable polymer), it was concluded that with comparable strut tissue coverage, the MiStent demonstrated a superior efficacy for preventing neointimal formation in the serial comparative OCT analysis of the MiStent versus the XIENCE at six months. However, this difference in %VO (Volume Obstruction) decreased at 24 months, rendering the difference in neointimal at 24 months insignificant⁸. Also another study supports the early safety and efficacy with MiStent's sirolimus-eluting bioabsorbable polymer-coated stent when compared to Xience's everolimus-eluting permanent polymer-coated stent in a sizable all-comers population at a longer term follow-up⁹. Studies also proved that Treatment with BP-SES compared to DP-EES was linked to a significantly decreased rate of 12-month target lesion failure in the ACS population. These results suggest that ACS patients receiving percutaneous coronary intervention should be treated with an ultrathin strut BP-SES¹⁰. The outcomes of one of the study supported that bioresorbable polymer stent's good long-term safety and effectiveness in this high-risk patient population¹¹.

- B. Several studies have shown that Everolimus Stent is clinical better than other Drug Eluting stents like Sirolimus and Paclitaxel stents. One of the study revealed that in patients with 2- to 3-vessel MV-CAD (Multivessel - Coronary Artery Disease), there was a decreased in-stent late loss with the EES XIENCE V compared with the PES TAXUS Libertè at 9 months and a low major adverse cardiac event rate at 1

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year¹². Further, as compared to PES, coronary implantation of EES resulted in lower rates of target lesion failure, myocardial infarction, ischemia-driven target lesion revascularization, and all-cause mortality in this large dataset with a 3-year follow-up. It is necessary to conduct more research to characterize any potential relationships between the type of stent, diabetes, and vascular¹³. EES compared to PES had long-lasting advantages in composite safety and effectiveness metrics as well as all-cause mortality at 5 years after therapy. Additionally, late follow-up showed no degradation in effect over time despite the absolute difference in TLR between devices being consistent¹⁴. Studies also proved that EES has better clinical outcomes than PES, mostly as a result of a decline in the incidence of TVR brought on by ischemia¹⁵. One of the studies suggested that 2nd generation Drug eluting stents (DES-2s) are better than Bare Metal Stents (BMS), with TVR rate ratio, DES-2s are significantly more efficient than BMSs (i.e., 0.29 to 0.62 in more recent meta-analyses). With the exception of the cobalt-chromium everolimus DES, findings are infrequently noteworthy, while DES-2s appear to cause fewer fatalities and in-stent thrombosis than BMSs. Myocardial infarction rate ratio is consistently in favor of DES-2s and is frequently considerable¹⁶. Also, the safety and efficacy profile of second-generation DES appears to be superior to that of BMS, with decreased rates of MACE, MI, TLR, and stent thrombosis¹⁷. In comparison to SES and PES, CoCr-EES showed higher strut coverage, less inflammation, less fibrin deposition, and less LST/VLST during human postmortem investigation. However, the reported incidences of neo atherosclerosis and fracture-related adverse pathologic events were equal in these devices, demonstrating the need for rigorous long-term follow-up even after the implantation of CoCr-EES¹⁸. However, studies suggested that both EES and ZES (Zotarolimus Eluting Stent) implantation were efficacious and safe in lowering neointima volume and late loss with a comparable rate of MACEs irrespective of cardiovascular risk factors in patients with first- and second-generation DES restenosis¹⁹.

- C. Several studies demonstrated that everolimus stents has better clinical outcomes. Patients who had everolimus-eluting stents implanted did not experience an increase in the composite outcomes of cardiac mortality, myocardial infarction, stroke, or TIMI significant hemorrhage at 1 year compared to those who had 12-month DAPT (Dual Antiplatelet Therapy)²⁰. One of the study conducted on Indian patients proved that the use of XIENCE V in "real world" Indian patients was associated with very low

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clinical event rates up to three years of follow up, despite the high risk population with coronary artery disease²¹.

- D. Safety and Effectiveness of Equivalent device (Evermine 50) is already proven through several clinical studies which are discussed in Clinical Evaluation Report.

Conclusion: As discussed in the section above, biodegradable polymer stent with ultrathin struts have greater clinical advantages over the durable polymer stents. The subject device also uses biodegradable polymer and has ultrathin strut ($65 \pm 10 \mu\text{m}$), hence the same benefits are expected from the subject device too. Further the section above, also states that Everolimus Stents are better than other DES and BMS. The subject device too is Everolimus stent, so these benefits are also expected in subject device. Further the safety and performance of Equivalent device is already proved through several clinical studies, hence the same safety and performance is expected for subject device too. The clinical experience with subject device – from customer complaints, product feedback and PMCF study results, clearly shows that subject device is also safe and effective for its intended use.

5.3.2 Post Market Clinical Follow up (PMCF) Study

The Post Market Clinical Follow up Study for the subject device is ongoing. The details of the study along with its results are presented in this section.

Synopsis of PMCF study

Study Title	A real world, all comer, prospective, multicenter, Clinical study to evaluate the safety and performance of the Eterna -Everolimus Eluting Coronary Stent System (EEECSS-Eterna) implanted as a part of standard clinical practice in India-ALCHEMY
Study Protocol Number, Version and date	IHPL-CT-001-23, V_1.0.0 dated 12-06-2023
Study Centre	Multi Center (25-30 sites)
Number of patients	1000 patients

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Sponsor	Innovation Healthcare Pvt. Ltd. Plot No. 105, Solitaire Industrial Park, Village Dehmikalan, The. Sanganer, Jaipur – 303007, Rajasthan (India)
Investigational device	Eternia Everolimus Eluting Coronary Stent System
Intended use	The device is intended for use in the patient eligible for Percutaneous Transluminal Coronary Angioplasty (PTCA). Everolimus Eluting Coronary Stent System is indicated for improving coronary luminal diameter in patients with symptomatic ischemic disease due to discrete <i>de novo</i> lesions of length ≤ 48 mm in native coronary arteries with a reference vessel diameter of 2.00 to 4.50 mm.
Study Objectives	<u>Primary objective:</u> To evaluate the safety and performance of EEECSS at 1 year follow up indicated by the Device Oriented composite end point (DoCE) defined as Cardiac Death, target vessel Myocardial infarction (TV-MI) and clinically driven target lesion revascularization (TLR) <u>Secondary objectives:</u> To estimate incidence of EEECSS patient oriented composite end point (PoCE) defined as all cause death, any myocardial infarction and any revascularization, Target vessel failure (TVF), Target Vessel related Myocardial Infarction (TV-MI), Non -Target Vessel related Myocardial Infarction and individual components of composite end points at 30 days, 180 days, 270 days and 1 year. For subset analysis the objective is to measure Late Lumen Loss on QCA and percentage of covered struts on OCT at 9 months.
Study Design	This will be a prospective, open label, non-randomized, single-arm, multi-centre, Clinical study enrolling patients with ischemic heart disease attributable to native coronary artery stenosis that are amenable to treatment by PCI with stenting.

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Study Population	Patients of coronary Artery Disease (CAD) eligible for Eternia Everolimus Eluting Coronary Stent Implantation.
No. of Subjects & duration	A total of 1000 subjects will be enrolled for 1 year of study duration
Method of clinical investigation	Refer PMCF plan (IHPL-CT-001-23, V_1.0.0 dated 12-06-2023)
The results of the clinical investigation	The study is going. However, interim results of first 100 patients who underwent percutaneous coronary intervention with implant of one or more Eternia Everolimus Eluting Stents are obtained. The early results of this post marketing study clearly demonstrate 100 % device success and 100 % procedure success. Hence, the device is safe and effective as per indicated in patient undergoing Percutaneous Transluminal Coronary Angioplasty (PTCA).
The conclusion	The study is ongoing

5.4 An overall summary of the clinical performance and safety

A minimally invasive procedure, percutaneous transluminal coronary angioplasty (PTCA), also known as percutaneous coronary intervention (PCI), is used to open blocked or stenosed coronary arteries and restore blood flow to the myocardium. Lipid-rich plaque in the arteries causes the blockages, which reduce blood flow to the myocardium²².

Atherosclerosis refers to the buildup of lipid-rich plaque in the arteries. At the point when atherosclerosis influences the coronary arteries, the problem is known as coronary artery disease (CAD). Patients with CAD normally present with exertional chest pain, or with dyspnea with exertion. In acute myocardial infarction, there is plaque rupture with platelet aggregation, and acute thrombus formation, which results in a sudden occlusion of the coronary artery. These patients present with acute chest heaviness, diaphoresis, and nausea. Urgent PTCA is often required to limit myocardial damage²².

The most common technique for coronary artery revascularization is percutaneous coronary intervention. Initial balloon angioplasty procedures were ineffective due to the high rates of abrupt vascular closure and

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clinical restenosis caused by coronary dissections, arterial recoil, and neointimal development. Vascular dissections were stabilized, and arterial recoil was reduced with the invention of coronary stents, but neointimal buildup remained an issue, leading to the development of in-stent restenosis (ISR) in 20%–30% of instances. For the purpose of reducing neointimal development, drug-eluting stents (DESs) were created to release antiproliferative drugs at the site of artery damage. Although delayed re-endothelialization and stent thrombosis are still significant obstacles, DESs have steadily improved outcomes following percutaneous coronary intervention²³

The stainless-steel stents from the first generation had thicker coatings. Even if they outperformed BMS in terms of performance, there was still space for development. The second generation of stents then appeared, with better metal platforms with thinner struts and thin coatings. Paclitaxel and Sirolimus were replaced with Zotarolimus and Everolimus. The best-performing stents were these, although there was a problem with the permanent coating, which persisted over the stent surface even after full drug elution and began to produce problems in longer-term tests. The third generation of drug eluting stents, used biodegradable polymer. Thus providing the enhanced DES²⁴.

The clinical benefits associated with the use of the device includes:

- It attributes to reduced rate of restenosis as compared to Bare Metal Stent (BMS) other Drug Eluting Stents (DES), which is < 8%
- It provides reduced rate of Target lesion revascularization (TLR) as compared to other DES, which is < 7%
- It provides reduced rate of Stent thrombosis as compared to other DES, which is < 5%
- It contributes to fewer myocardial infarctions (MI), at two years follow up as compared to other DES, which is < 4%

These benefits are supported by clinical evidences, which are presented in Clinical Evaluation Report.

The device is permanent implant and will remain in body for the entire life of the patient.

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Risk management activities were done in line with requirements/guidance of applicable standards (e.g. ISO 14971). All the possible hazards, including clinical risks, that could have impact on the safety of the device were clearly determined. The risks have been identified from all the stages of device life cycle. The identified risks were mitigated by providing the control measures. All the risks were accepted after placing the control measures. After review of the mitigation steps, it was concluded that no additional risks were induced. It is concluded that, based on the risk/benefit analysis, the benefit of the device is higher than the residual risk. The overall risk is therefore at the minimum possible level and the overall risk of the device is within acceptable limits. Clinical risks have also been derived from the product feedback and complaints of the device under evaluation. After evaluation of product feedback, no clinical risks were found. The benefit-risk ratio is more than 1, indicating that the benefits outweigh the risks.

5.5 Ongoing or planned post-market clinical follow-up

Post-market clinical follow-up for Everolimus Eluting Coronary Stent System will be performed at least yearly and will include analyses of real-world evidence by performing a literature search, screening of device registers for clinical data, as well as analysis of all complaints, customer/market feedback, vigilance. The Summary of Safety and Clinical Performance will be updated with information from the post-market clinical follow-up, if this is needed to ensure that any clinical and/or safety information described in this document remains correct and complete.

6 Possible diagnostic or therapeutic alternatives

Following treatment options are currently available for the treatment of Coronary Artery Disease.

6.1 Angioplasty

Angioplasty is a nonsurgical procedure that opens blocked or narrowed coronary arteries. This procedure also is called percutaneous transluminal coronary artery angioplasty, or PTCA. A thin, flexible tube with a balloon or other device on the end is threaded through a blood vessel to the narrowed or blocked coronary artery. Once in place, the balloon is inflated to compress the plaque against the wall of the artery. This restores blood flow through the artery. During the procedure, the doctor may put a small mesh tube called a stent in the artery. The stent helps prevent blockages in the artery in the months or years after angioplasty.

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6.2 Coronary Artery Bypass Grafting (CABG)

An example of surgery is CABG. In CABG, your clogged coronary arteries are bypassed (i.e., moved around) using arteries or veins from other parts of your body. With CABG, your heart's blood flow can be enhanced, your chest pain can be reduced, and a heart attack may be avoided.

6.3 Cardiac Rehabilitation

Cardiac rehabilitation for angina or after CABG, angioplasty, or a heart attack may be prescribed.

Rehab has two parts:

- Exercise training.
This part helps you learn how to exercise safely, strengthen your muscles, and improve your stamina. Your exercise plan will be based on your personal abilities, needs, and interests.
- Education, counseling, and training.
This part of rehab helps you understand your heart condition and find ways to reduce your risk for future heart problems. The rehab team will help you learn how to cope with the stress of adjusting to a new lifestyle and deal with your fears about the future.

7 Suggested Profile and Training of the Users

The subject device may only be used by Professional and Registered Cardiologist/doctors trained and experienced in the PTCA procedure.

8 Reference to any standards applied

List of Applicable Standards (Harmonized Standards)

Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
1.	EN ISO	Medical device - Quality	The organization is involved	Full

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Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
	13485:2016/A11:2021	management systems - Requirement for regulatory purposes	in manufacturing of medical devices and its QMS is certified as per EN ISO 13485:2016, hence, this standard is applicable	
2.	EN ISO 20417:2021	Information supplied by the manufacturer of medical devices	Being the manufacturer of medical devices, this standard is followed to provide the information of medical device to the users.	Full
3.	EN 62366:2008	Medical devices — part 1: application of usability engineering to medical devices	The concerned product is medical device; hence this standard is followed for usability engineering applications of the concerned product	Full
4.	EN ISO 25539-1:2017	Cardiovascular implants — endovascular devices — part 1: endovascular prostheses	As the concerned product is cardiovascular implant – the stent, hence, these standards are applicable.	Full
5.	EN ISO 25539-2:2020	Cardiovascular implants — endovascular devices — part 2: vascular stents		Full
6.	EN ISO 10993-1:2020	Biological evaluation of	It is essential to check the	Full

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Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
		medical devices - Part 1. Evaluation and testing	biocompatibility of the medical device before placing it in the market. ISO 10993 series of standards gives the test requirements and method for biocompatibility testing of medical devices. Hence these standards are applicable.	
7.	EN ISO 10993-3:2014	Biological evaluation of medical devices - Part 3. Tests for genotoxicity, carcinogenicity and reproductive toxicity		Full
8.	EN ISO 10993-4:2017	Biological evaluation of medical devices – Part 4: Selection of tests for interactions with blood		Full
9.	EN ISO 10993-5:2009	Biological evaluation of medical devices - Part 5. Tests for in vitro cytotoxicity		Full
10.	EN ISO 10993-6:2016	Biological evaluation of medical devices - Part 6. Tests for local effects after implantation		Full
11.	EN ISO 10993-7:2008/A1:2022	Biological evaluation of medical devices - Part 7: Ethylene oxide sterilization residuals		Full
12.	EN ISO 10993-	Biological evaluation of		Full

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Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
	10:2023	medical devices - Part 10. Tests for irritation and delayed-type hypersensitivity		
13.	EN ISO 10993-11:2018	Biological evaluation of medical devices - Part 11. Tests for systemic toxicity		Full
14.	EN ISO 10993-12:2021	Biological evaluation of medical devices - Part 12: Sample preparation and reference materials		Full
15.	EN ISO 15223-1:2021	Medical devices - Symbols to be used with medical device labels, labelling and information to be supplied - Part 1: General requirements	As the concerned device is a Medical device, this standard is applicable	Full
16.	EN ISO 11737-1:2018+A1:2021	Sterilization of health care products – Microbiological methods - Part 1: Determination of a population of microorganisms on products	Being a medical device, it is a health care product and bioburden estimation on the product is carried out, hence this standard is applicable	Full

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Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
17.	EN ISO 11737-2:2020	Sterilization of medical devices – Microbiological methods - Part 2: Test of sterility performed in the definition, validation and maintenance of a sterilization process	The concerned product is supplied sterile, hence to check whether the sterilization is achieved, sterility testing is performed. Thus, this standard is applicable.	Full
18.	EN ISO 11607-1:2020/A1:2023	Packaging for terminally sterilized medical devices - Part 1: Requirement for material, sterile barrier system and packaging system	The concerned product is terminally sterilized; hence these standards are applicable	Full
19.	EN ISO 11607-2:2020/A1:2023	Packaging for terminally sterilized medical devices - Part 2: Validation requirement for forming, sealing and Assembly process		Full
20.	EN ISO 14155:2020	Clinical investigation of medical devices for human subjects - Good clinical practice	Being a medical device, this standard is applicable for clinical investigation	Full
21.	EN ISO 11138-2:2017	Sterilization of healthcare	As the sterilization process	Full

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Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
		products - Biological indicators - Part 2: Biological indicators for ethylene oxide sterilization processes	is done via Ethylene oxide and biological indicators are used during this process, hence this standard is applicable	
22.	EN ISO 11135:2014/A1:2019	Sterilization of health-care products — Ethylene Oxide — Requirements for the development, validation and routine control of a sterilization process for medical devices — Amendment 1: Revision of Annex E, Single batch release	The concerned product is supplied in sterile form and the mode of sterilization is EO sterilization. Further, this is new standard and this new version is not available in Harmonized standards, hence this standard is applicable, considering state of art.	Full
23.	EN ISO 14644-1:2015	Cleanrooms and associated controlled environments - Part 1: Classification of air cleanliness by particle concentration	The concerned product is manufactured in clean room environment and these standards are not available in harmonized standards, hence these standards are applicable.	Full
24	EN ISO 14644-2:2015	Cleanrooms and associated controlled		Full

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Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
		environments - Part 2: Monitoring to provide evidence of cleanroom performance related to air cleanliness by particle concentration		
25	EN ISO 14644-3:2019	Cleanrooms and associated controlled environments - Part 3: Test methods		Full
26	EN ISO 14644-4:2022	Cleanrooms and associated controlled environments - Part 4: Design, construction and start-up		Full
27	EN ISO 14644-5:2004	Cleanrooms and associated controlled environments - Part 5: Operations		Full
28	EN ISO 14644-7:2004	Cleanrooms and associated controlled environments - Part 7: Seperative devices (clean air hoods, gloveboxes,		Full

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Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
		isolators and mini-environments)		
29	EN ISO 14971:2019/A11:2021	Medical devices - Application of risk management to medical devices	The concerned product is medical device and this standard gives the risk management aspects of medical devices, this standard is applicable	Full
	EN ISO 11138-1:2017	Sterilization of health care products — Biological indicators — Part 1: General requirements	As the medical device is sterilized and biological indicators are used during this process, this standard is applicable.	Full

List of Applicable Standards (Non-Harmonized Standards)

Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
1.	ISO 14698-1:2003	Cleanrooms and associated controlled environments -- Biocontamination control -- Part 1: General principles	The concerned product is manufactured in clean room environment and these standards are not available in harmonized standards, hence these standards are applicable.	Full

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		and methods		
2.	ISO 14698-2:2003/Cor 1:2004	Cleanrooms and associated controlled environments -- bio contamination control -- Part 2: Evaluation and interpretation of bio contamination data		Full
3.	ISO/TR 24971:2020	Medical devices — Guidance on the application of ISO 14971	The concerned product is medical device and this standard gives the risk management aspects of medical devices, this standard is applicable	Full
4.	USP 35 General chapter <71>	Sterility tests	As the product is supplied sterile in the market, sterility testing is performed on finished product. This testing is performed as per this standard. Hence this standard is applicable.	Full
5.	USP 35 General chapter <85>	Bacterial endotoxins test	As the product is claimed as non- pyrogenic, BET testing is performed to prove that product is non- pyrogenic. BET test is performed using this standard; hence this standard is applicable.	Full
6.	ASTM F1980-16	Standard Guide for	Stability study of the concerned	Full

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		Accelerated Aging of Sterile Barrier Systems for Medical Devices	medical device follows this standard; hence this standard is applicable.	
7.	ASTM D4169-16	Standard Practice for Performance Testing of Shipping Containers and Systems	The transport validation is performed as per this standard; hence this standard is applicable.	Full
8.	ISTA 2A-2011	Packaged-products weighing 150 lbs. (68kg) or less.	This standard too is used for transport validation; hence this standard is applicable.	Full
9.	Ph. Eur (2019) 10 th edition	European Pharmacopoeia	This standard is used for bioburden testing of the device. Hence this standard is applicable	Full

ICH Guidelines

Sr. No.	Document No.	Title	Rational	Compliance Information (Full/Partial)
1.	ICH Q1A (R2)-2003	Stability Testing of New Drug Substances and Products	As Product contains drug – Everolimus, this standard is applicable.	Full

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List of Applicable Regulation

Directive	Title	Rational for Selection	Compliance Information (Full/Partial)
EU MDR 2017/745	European Union Medical Device Regulation	This regulation is applicable to Medical Devices and as our product is medical device, this directive is used for complying with its requirements and for obtaining CE certification	Full

9 Summary of the safety and clinical performance of the device intended for patients

A summary of the safety and clinical performance of the device intended for patients, is not applicable as the device is for Cardiologist / Doctors use only.

10 Revision history

SSCP revision number	Date issued	Change description	Revision validated by the Notified Body
00	15-01-2024	Initial version	Date: not yet Validation language: English

11 References

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